ALEPH Charged-Particle Cumulants

Status Report

Yen-Jie Lee

Electron-Positron Alliance / MIT HIG

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Current Status

- Built a standalone ROOT/C++ cumulant workflow using charged particles in archived ALEPH LEP1 data.
- Current data input is the merged LEP1 1992–1995 StudyMult sample with **3,050,610 events**.
- Results are detector-level and available for beam-axis and thrust-axis azimuths.
- Outputs now include inclusive, MC-only, rapidity-gap, two-subevent, and MC-comparison plots.

Important caveat: no converted 1991 ROOT sample was found in the checked merged StudyMult directories, so the current data result is 1992–1995 rather than complete 1991–1995 LEP1.

Samples and Selection

Data

- `LEP1Merged_20200611.root`
- Tree: `t`, StudyMult fixed-array format
- 1992: 551,474 events
- 1993: 538,601 events
- 1994: 1,365,440 events
- 1995: 595,095 events

MC baseline

- `LEP1MCMerged.root`
- 771,597 reconstructed entries
- 1994-only baseline comparison

Track/event settings

- StudyMult charged flags: `pwflag = 0,1,2`
- charged-particle threshold: $p_T > 0.4$ GeV
- multiplicity bins: $[0, 10), \ldots, [40, \infty)$

Analysis Workflow

- Q-cumulants are accumulated for harmonic $n = 2$: $\langle 2 \rangle$, $\langle 4 \rangle$, $\langle 6 \rangle$, $\langle 8 \rangle$.
- Derived observables: $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, $v_2\{8\}$.
- Each production is split into 16 chunks; statistical errors use delete-one-chunk jackknife after recomputing the nonlinear cumulant values.
- Internal self-test compares ordered-correlator implementation with brute-force distinct-particle sums.
- Current plots compare data points to the 1994 MC baseline, with MC drawn as histograms or bands where applicable.

Cumulant Definitions

- Within each $N_{trk}^{offline}$ bin, event sums are combined into weighted correlators before forming cumulants:

$$\langle\langle 2k\rangle\rangle = \frac{\sum_{\text{events}} \mathrm{Num}_{2k}}{\sum_{\text{events}} \mathrm{Den}_{2k}}, \qquad \mathrm{Num}_{2k} = \mathrm{Re}\sum_{\text{distinct}} e^{2i(\phi_1+\cdots+\phi_k-\phi_{k+1}-\cdots-\phi_{2k})}.$$

$$\begin{aligned}
c_2\{2\} &= \langle\langle 2\rangle\rangle, \\
c_2\{4\} &= \langle\langle 4\rangle\rangle - 2\langle\langle 2\rangle\rangle^2, \\
c_2\{6\} &= \langle\langle 6\rangle\rangle - 9\langle\langle 4\rangle\rangle\langle\langle 2\rangle\rangle + 12\langle\langle 2\rangle\rangle^3, \\
c_2\{8\} &= \langle\langle 8\rangle\rangle - 16\langle\langle 6\rangle\rangle\langle\langle 2\rangle\rangle - 18\langle\langle 4\rangle\rangle^2 \\
&\quad + 144\langle\langle 4\rangle\rangle\langle\langle 2\rangle\rangle^2 - 144\langle\langle 2\rangle\rangle^4.
\end{aligned}$$

Inclusive denominators are $N(N-1)\cdots(N-2k+1)$; two-subevent denominators are $N_A(N_A-1)\cdots(N_A-k+1)\,N_B(N_B-1)\cdots(N_B-k+1)$.

Extracting $v_2\{2k\}$

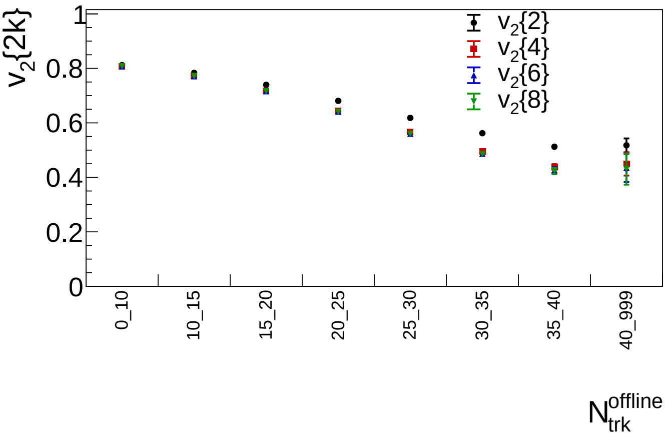
$$v_2\{2\} = \sqrt{c_2\{2\}},$$
$$v_2\{4\} = \left(-c_2\{4\}\right)^{1/4},$$
$$v_2\{6\} = \left(c_2\{6\}/4\right)^{1/6},$$
$$v_2\{8\} = \left(-c_2\{8\}/33\right)^{1/8}.$$

Valid real roots require

$$c_2\{2\} \geq 0, \quad c_2\{4\} < 0,$$
$$c_2\{6\} > 0, \quad c_2\{8\} < 0.$$

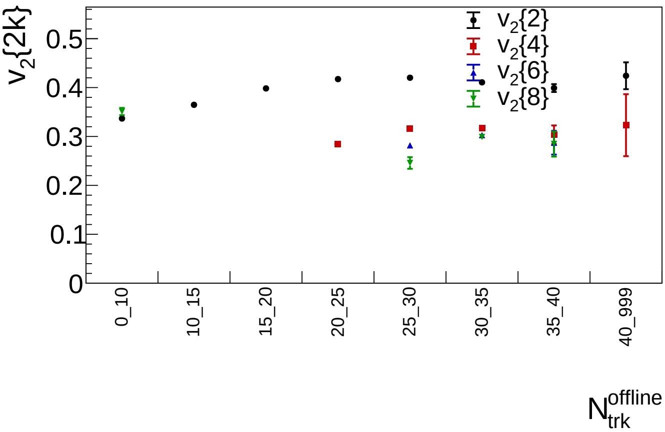
- The two-subevent version uses the same cumulant algebra, but each $\langle\langle 2k \rangle\rangle$ samples k tracks from $\eta > 0$ and k tracks from $\eta < 0$.
- Statistical errors are evaluated with 16 delete-one-chunk jackknife samples after recomputing the nonlinear $c_2\{2k\} \rightarrow v_2\{2k\}$ transformation.
- A $v_2\{2k\}$ bin is not drawn if the central value or any jackknife sample fails the required real-root sign.

Inclusive Result: Beam Axis



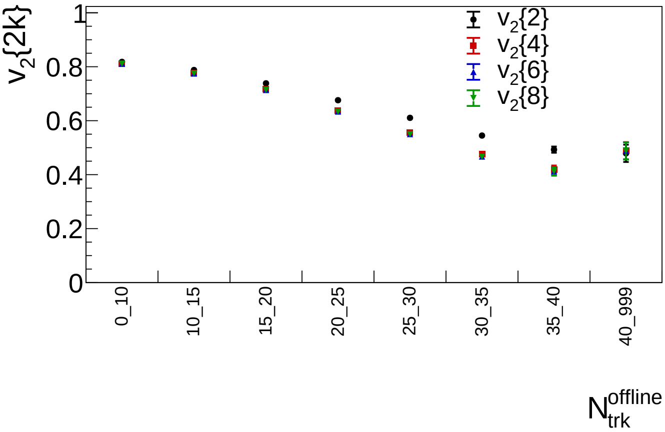
Merged LEP1 1992–1995 data; detector-level charged particles with $p_T > 0.4$ GeV.

Inclusive Result: Thrust Axis



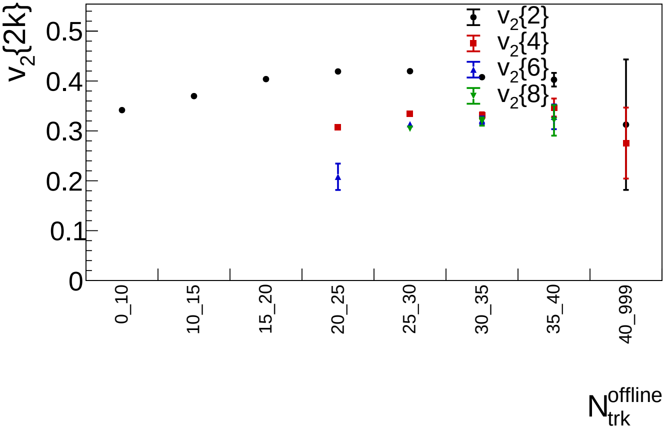
Same selected charged-particle multiplicity bins, with azimuth evaluated around the event thrust axis.

MC-Only Inclusive: Beam Axis



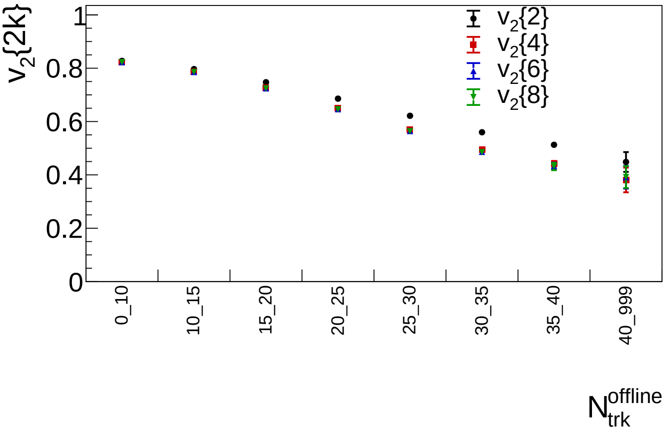
1994 reconstructed MC baseline; same charged-particle selection and lab multiplicity bins.

MC-Only Inclusive: Thrust Axis



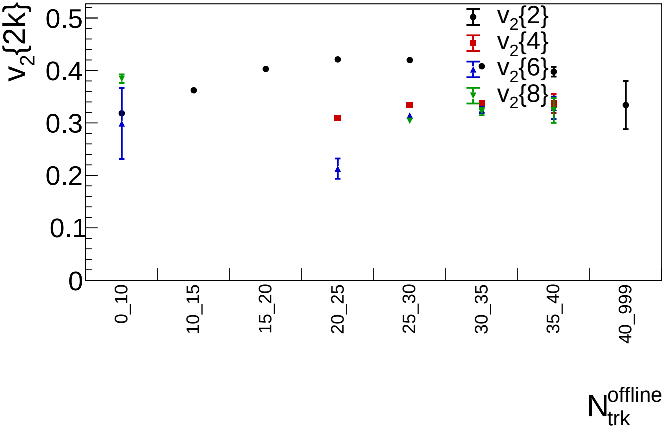
1994 reconstructed MC baseline with azimuth and pseudorapidity evaluated around the event thrust axis.

MC-Only Gen Level: Beam Axis



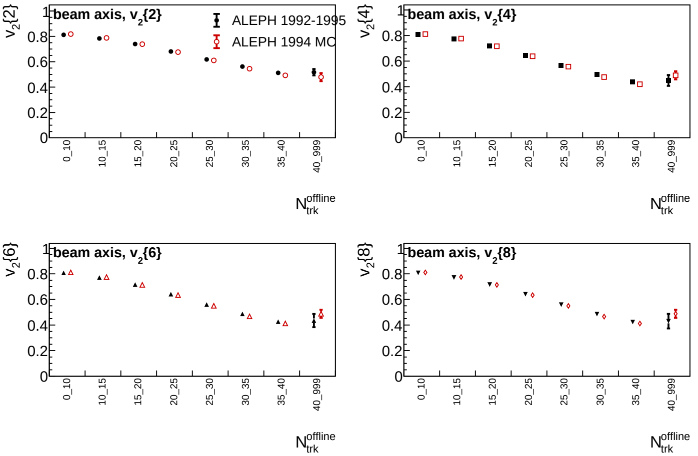
Generator-level charged particles from the *tgen* tree of `LEP1MCMerged.root`; same $p_T > 0.4$ GeV selection and $N_{trk}^{offline}$ binning.

MC-Only Gen Level: Thrust Axis



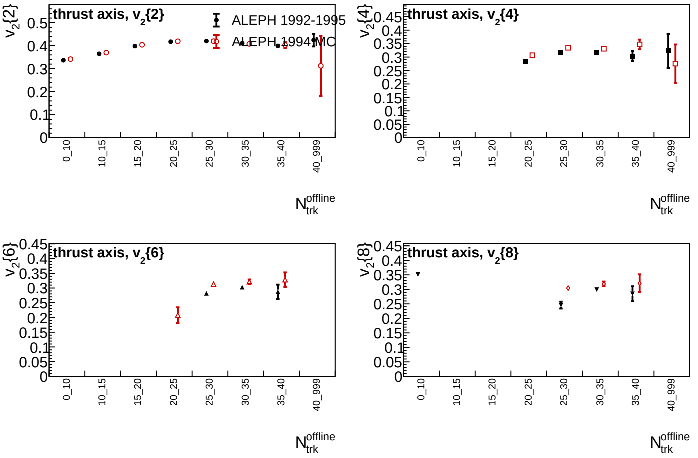
Generator-level MC baseline with azimuth and pseudorapidity evaluated around the event thrust axis.

Data/MC Comparison: Beam Axis



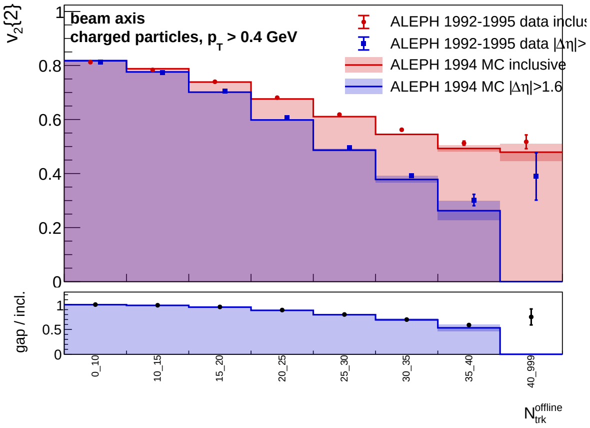
Data: LEP1 1992–1995; MC: 1994 reconstructed baseline.

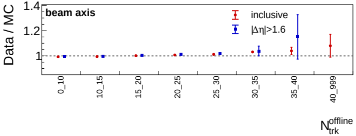
Data/MC Comparison: Thrust Axis



Data: LEP1 1992–1995; MC: 1994 reconstructed baseline.

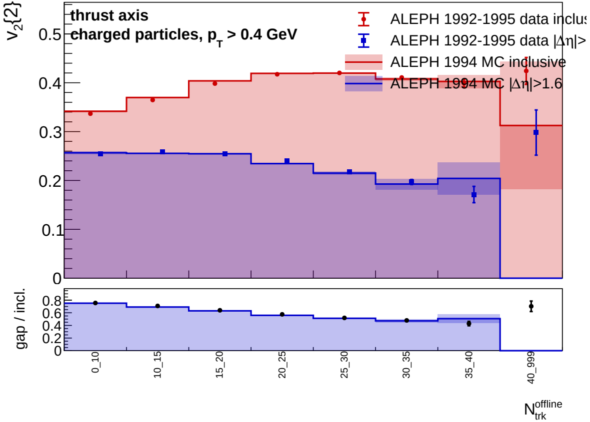
Rapidity Gap: $|\Delta\eta| > 1.6$, Beam Axis

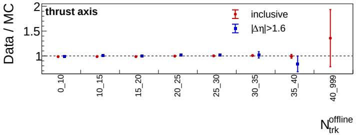




- Looser than the $|\Delta\eta| > 2.0$ study.
- Ratio panel tracks data/MC for inclusive and gap observables.
- Largest uncertainty remains in the high-multiplicity tail.

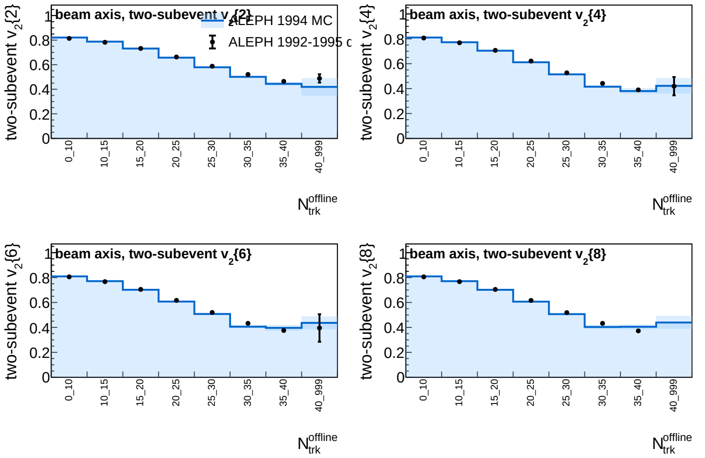
Rapidity Gap: $|\Delta\eta| > 1.6$, Thrust Axis





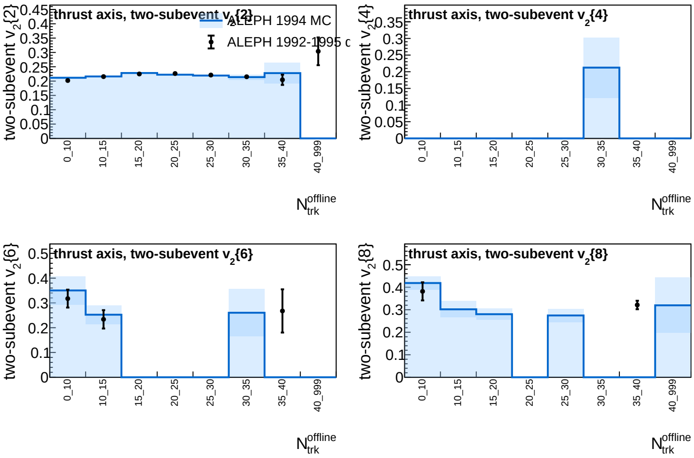
- Gap is evaluated in pseudorapidity around the thrust axis.
- Suppression is stronger than in the beam-axis view.
- Data and MC follow the same broad multiplicity trend.

Two-Subevent Cumulants: Beam Axis



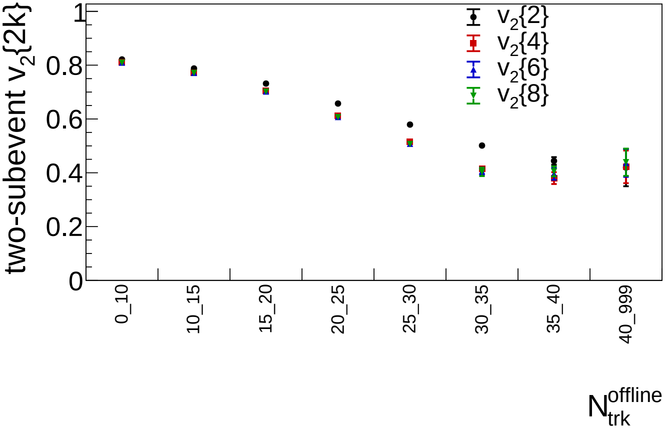
Two-subevent result samples k particles from $\eta > 0$ and k particles from $\eta < 0$ for $v_2\{2k\}$.

Two-Subevent Cumulants: Thrust Axis



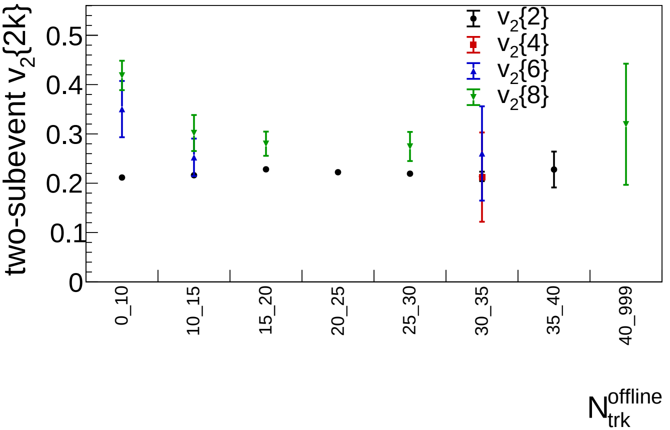
Two-subevent result samples k particles from $\eta > 0$ and k particles from $\eta < 0$ for $v_2\{2k\}$.

Two-Subevent MC Only: Beam Axis



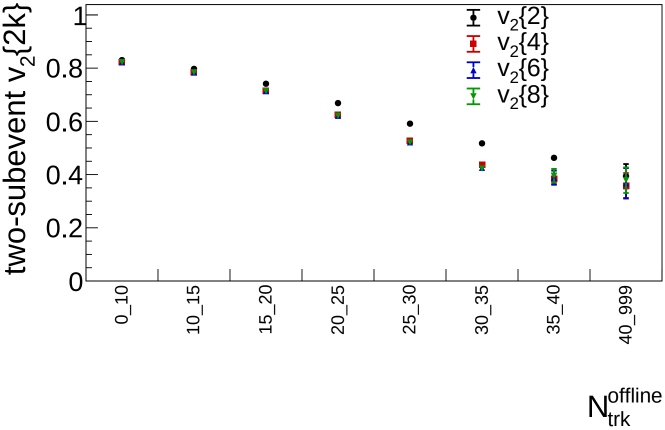
1994 reconstructed MC baseline; two-subevent cumulants only, using $\eta > 0$ and $\eta < 0$ subevents.

Two-Subevent MC Only: Thrust Axis



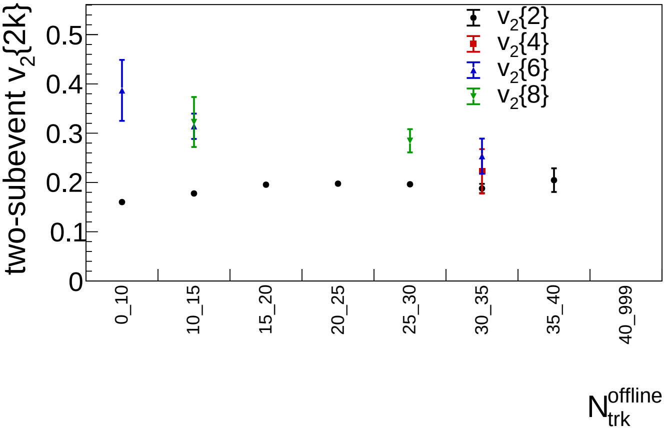
MC-only view of the sign-instability seen in the thrust-axis two-subevent higher-order cumulants, using thrust-axis $\eta > 0$ and $\eta < 0$ subevents.

Two-Subevent MC Gen Level: Beam Axis



Generator-level MC-only two-subevent cumulants from the *tgen* tree, with k particles sampled from $\eta > 0$ and k from $\eta < 0$.

Two-Subevent MC Gen Level: Thrust Axis



Generator-level MC-only two-subevent cumulants with the event thrust axis defining $\eta > 0$ and $\eta < 0$ subevents.

What We Learn So Far

- Beam-axis $v_2\{m\}$ values are large and decrease with multiplicity; high-order cumulants are stable in the populated bins.
- Thrust-axis values are smaller and high-order cumulants become sign-invalid in several low-multiplicity bins.
- Rapidity-gap selections reduce the inclusive two-particle coefficient, consistent with sizable short-range or jet-like contributions.
- The 1994 MC baseline broadly follows the data trends, but this is not yet a detector-corrected or systematic-uncertainty result.

Next Steps

- Add correction and systematic studies: tracking, event selection, axis definition, MC model dependence, and multiplicity migration.
- Stress-test rapidity-gap and two-subevent definitions against alternate split/gap choices.
- Expand MC comparisons beyond the current 1994 baseline where available.
- Keep the analysis note and reproducible run instructions synchronized with each production.

Current deliverable: GitHub and Overleaf contain the merged-data figures, CSV tables, note, and reproducible commands for the current detector-level status.